LARGE LANGUAGE MODELS IN HUMAN-MACHINE INTERACTION

Ömer Emre Mutlu

emre.mutlu@rwth-aachen.de

Abstract

This document serves as a template for the seminar write-up on Large Language Models in human-machine interaction. It will be replaced with the final abstract once the paper is complete.

Keywords: Large Language Models, Human-Machine Interaction, Seminar

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1 Introduction

In high-stakes environments such as aerospace, medicine, disaster response, military applications, and similar domains, the effectiveness and speed with which operators interact with the tools available to them are of critical importance. Traditionally, human–machine interaction (HMI) in such domains has been achieved through methods that have been field-tested over many years, such as user interfaces (UIs) utilizing buttons, joysticks, haptic-feedback sensors, and touchscreens. Depending on the application domain, specific norms and regulations govern the permissible interaction mediums for HMI user interfaces, including standards such as IEC 60204-1 for industrial electrical equipment and NASA-STD-3001 for aeronautical systems.[Int16][CWF23]

However, with the introduction of transformer encoder–decoder models and the subsequent popularization of large language models such as Google's BERT, OpenAI's GPT-3, and more recently GPT-5, a wide range of new possibilities has emerged in the field. Moreover, recent advancements in large language models in the areas of speech-to-speech and visual language models have created additional and particularly interesting opportunities.

This paper presents a literature review of recent advancements in relation to human–machine interaction within the context of safety-critical working environments.

2 Background

Before proceeding with the analysis of domain-specific work, a brief introduction to the involved technologies is necessary to provide the reader with sufficient background and contextual understanding. Accordingly, this section presents a concise overview of the relevant technologies.

In a human–machine interaction (HMI) system, the human can be regarded as a component equipped with four primary input–output channels: vision, hearing, touch, and motor activity (movement) [DFAB04]. Among these channels, vision and hearing are most commonly addressed in contemporary AI-driven interaction systems and are therefore particularly relevant in the context of large language model (LLM)–based interfaces. For the visual modality, current approaches primarily involve text-based language models and vision–language models that process textual input alongside visual information. For the auditory modality, interaction is typically realized through audio-based processing pipelines that combine speech recognition and speech synthesis with language models. The subsequent subsections further examine these modalities in more detail.

2.1 Text-Based LLM Interfaces

Since the introduction of the Transformer architecture in "Attention Is All You Need", text-based content generation and processing have become the central foundation of modern natural language processing (NLP).[VSP+17] In this context, large language models are a family of neural language models trained on vast amounts of textual data and capable of performing a wide range of natural language processing tasks, such as generating context-aware responses based on user-provided text prompts.[BMR+20] To utilize this family of models as a user interface, the model must be capable of understanding domain-specific context and of acting upon user prompts, rather than merely generating text responses. Several methods exist to enable this, a brief summary of these approaches is provided below.

Fine Tuning

Fine-tuning is the process of further training a pre-trained model on task- or domain-specific data by updating its existing parameters or, in some approaches, by introducing additional trainable parameters.[Mos23] By further training the model on a domain-specific dataset, for example technical data on extraterrestrial habitats, a base model such as GPT-5 can acquire contextual knowledge of the application domain and generate its responses accordingly.

However, a significant downside of this approach is the increased risk of model hallucinations. Hallucinations refer to the phenomenon in which a model generates incorrect or fabricated information in response to a given prompt.[GYA+24] Such behavior can pose a significant risk in safety-critical environments.

Retrieval Augmented Generation

Retrieval-Augmented Generation (RAG) introduces relevant external information to the language model without modifying the model's internal parameters.

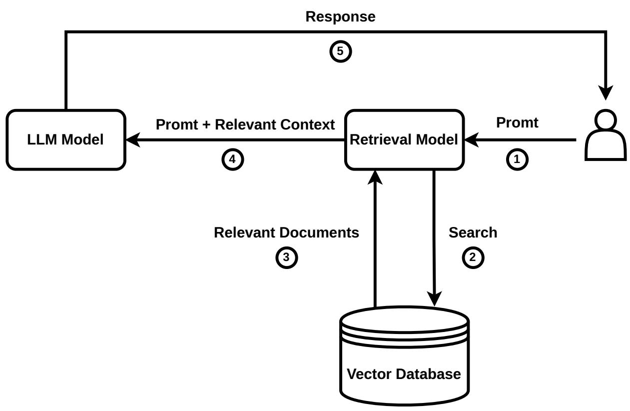

Figure 1: High-level RAG workflow that injects retrieved documents into the prompt instead of altering model weights.

As shown in Figure 1, the RAG workflow begins with a user prompt; however, instead of passing the prompt directly to the LLM, a similarity search is first performed over context-relevant information stored as embeddings in a vector database. The retrieved context is then provided together with the prompt to the LLM for grounded response generation.[BMR+20] Recent studies have shown this to be a more reliable method for context retrieval compared to fine-tuning.[OBME24]

However, the risk of hallucinations remains non-zero even with RAG; nevertheless, with well-designed context selection and retrieval strategies, this probability is significantly reduced, as demonstrated in numerous studies [JLF+24].

Tool Use / Function Calling

Tool calling, also referred to as function calling, is the ability of a large language model to invoke external functions by interpreting user context and generating structured Application Programming Interface (API) calls with the relevant parameters extracted from user prompts, as introduced by Schick et al. [SDYD+23].

In a practical setting, this means that the model generates structured tool calls, which are then used by a service layer to invoke HTTP requests to downstream services for task execution, as illustrated in Figure 2. By utilizing this method, the model can interact with external systems to trigger the execution of real-world functions.

In the context of human–machine interaction, this implies that a user can issue natural-language prompts to control robots and machines to execute a wide variety of tasks, as demonstrated by Vemprala et al. [VBBK23].

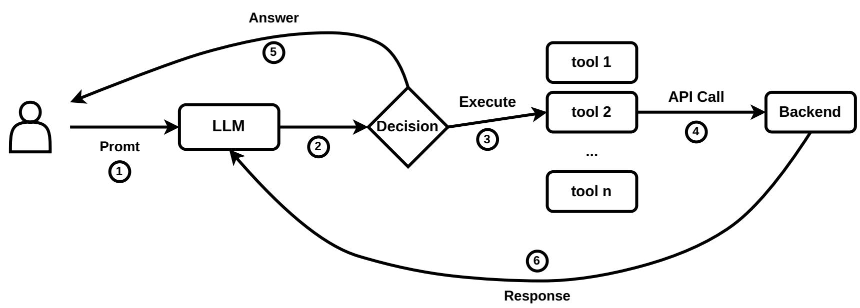

Figure 2: Conceptual workflow of tool (function) calling in a large language model.

Agentic Artificial Intelligence

What is commonly referred to as agentic AI is the combination of previously discussed methods to enable semi-autonomous decision-making systems. An AI agent typically integrates a large language model for reasoning and planning with tool-calling capabilities and memory in order to execute tasks without constant human supervision, requesting user input only when necessary to adjust its actions [YZY+23].

Such agents are currently most commonly used in programming-related tasks; a prominent example of an agentic AI system in this domain is OpenAI's Codex-based coding agent.

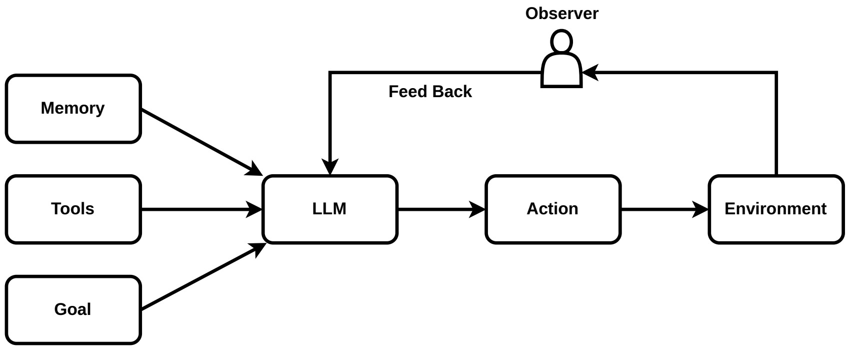

Figure 3: High-level agentic AI stack combining planning, memory, tool use, and execution loops.

With regard to the use of such systems in human–machine interaction, Kim et al. note that the application of agentic systems in this context remains underexplored. However, their user study reveals that current agentic systems are still prone to failure, which caused usability issues for some participants [KLM24]. The agents in their experiment exhibited persistent hallucinations, leading some participants to doubt their own knowledge. Additionally, in cases where a physical robot embodiment was used to represent the agent, participants reported an uncanny-valley-like sense of unease during task execution. Nevertheless, the field and the underlying technology are still at an early stage of development, and user experience is expected to improve over time. The application of such systems in domains such as deep-space exploration remains a particularly promising research direction, as discussed by Heinicke et al. [FSG+21]. However, in their current state, the suitability of such systems for safety-critical applications remains uncertain.

2.2 Audio Based Interfaces

A critical component of human–machine interaction is speech, which represents one of the most natural and powerful means of human communication. In many HMI application domains, the ability to translate spoken user input into corresponding machine actions is of particular importance. Accordingly, this section discusses two methods that are currently employed to enable speech-driven interaction in such systems.

Speech-to-Text, LLM and Text-to-Speech

As highlighted by Haeb-Umbach et al. and other previous work, until recently this architecture represented the primary approach for providing voice support in interactive human–machine interaction (HMI) applications [HUWN+]. The overall pipeline typically consists of a speech detection and preprocessing layer, which forwards the audio signal to a speech-to-text (STT) model, such as OpenAI's Whisper. The spoken input is thereby converted into text and subsequently processed by a large language model (LLM), which performs the reasoning and retrieval operations introduced in Section 2, such as retrieval-augmented generation (RAG). The LLM then produces a textual response, which is finally converted back into speech and presented to the user via a text-to-speech (TTS) system.

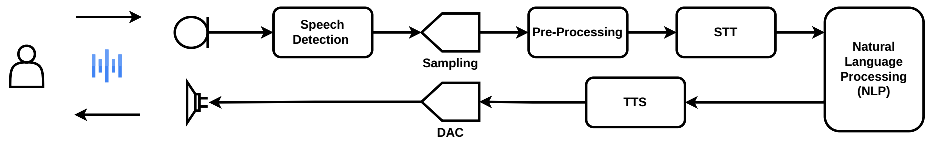

Figure 4: Signal chain for speech-to-text ingestion, LLM reasoning, and text-to-speech playback.

However, this pipeline also exhibits notable drawbacks. The sequential use of speech-to-text (STT), large language model (LLM) processing, and text-to-speech (TTS) synthesis can introduce perceptible latency. To mitigate this effect, some systems employ interaction design strategies such as playing short default utterances (e.g., "Let me think about that...") while the main response is being generated. Nevertheless, empirical user studies indicate that such waiting times can still negatively affect the naturalness and fluidity of human–machine interaction, particularly in conversational settings [FSG+21, KLM24].

On the positive side, such a configuration allows spoken input to be converted into textual form, which can facilitate integration with retrieval-augmented generation (RAG) systems and thereby support more efficient retrieval of contextually relevant information.

Speech-to-Speech LLMs

Speech

Vision-Language(-Action) Models

3 Domain Case Studies

3.1 Aerospace and Remote Science

Aerospace, Deep Sea wayages etc.

Psychological and Cognitive Impacts

3.2 Emergency and Environmental Response

3.3 Industrial Operations

4 Comparative Analysis and Metrics

5 Safety, Trust, and Deployment Constraints

6 Conclusion

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